LESSON PLAN · SCRATCH JUNIOR (K–2) BEGINNER: LITTLE CODERS

Lesson 8: Collect the Gems 💎

Module 3 | 30 minutes | Planning a path (algorithms) | KG–Grade 2 · ages 5–7

Objective: Plan a path to visit every gem and use a pick-up block at each one.

Success criteria

- I can plan a path to every gem.
- I can pick up gems.
- I can use a repeat for gems in a row.

Key vocabulary

- **collect**: Pick up and keep.
- **plan**: Decide the path first.
- **all**: Every one.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (30 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Gem hunt (5 min): paper gems on a floor grid; a child follows arrow cards to collect them.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Gem hunt: learners follow the steps.
5 min	**Perform**	Two gems, Gem row, On or next to?
3 min	**Probe**	Gem garden
4 min	**Wrap-up**	Exit ticket: What block do you use when you are on a gem?

Unplugged warm-up

Gem hunt (5 min): paper gems on a floor grid; a child follows arrow cards to collect them.

Teaching the idea

Algorithmic planning: decompose a goal (collect all) into a route, sequence moves and actions, use loops for regular patterns. Pick-up only works on the gem square (precise positioning).

Analogy: An Easter egg hunt with a plan.

Live demo: Trace two different routes on the board; count the moves each needs.

Questions to ask

Ask	Listen for
Where do you pick up?	On the gem.
Which gem first?	The closest.
Can a repeat help?	Yes, for gems in a row.

Differentiation

Support
Number the gems in the order to visit.

Stretch
Find the path with the fewest blocks.

Common misconceptions

Misconception	How to fix it
Walking past a gem collects it	Use the pick-up block on it.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

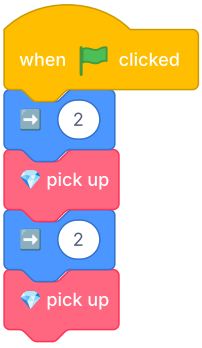
Exit ticket
What block do you use when you are on a gem?

Homework
Hide 3 toys at home and give someone arrow directions to find them.

Answer key — Kitabash activities

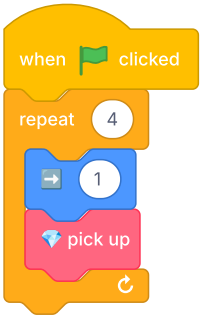
1. Two gems (perform · block code)

One solution:



2. Gem row (perform · block code)

One solution:



3. On or next to? (perform · multiple choice)

Answer: Nothing — it must be ON the gem

4. Gem garden (probe · block code)

One solution:



Answer key — worksheet

1. **Key words:** collect = Pick up and keep. · plan = Decide the path first. · all = Every one.
2. **On or next to?:** A. Nothing — it must be ON the gem — Move onto the gem first.
3. **Plan the gem path:** Any path that visits all three gems.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.